

Program : Diploma in Instrumentation Engineering	
Course Code : 3081	Course Title: Digital Circuits and Systems
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To provide a thorough knowledge in number system, sequential and combinational circuits.
- To develop digital circuits to solve real world problems.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Fundamentals of Digital		Fundamentals of electrical and electronics	2

Course Outcomes:

On completion of the course, the student will be able to

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the K map and Logic rules to simplify the Boolean expressions	11	Applying
CO2	Design the combinational circuits such as full adder, full subtractor, Multiplexer, gray code converter	13	Applying
CO3	Design the sequential logic circuits such as asynchronous and synchronous counter	14	Applying
CO4	Describe the working of ADC and DAC	5	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply the K map and Logic rules to simplify the Boolean expressions		
M1.01	Identify and apply, arithmetic and conversion operations on different number systems	4	Applying
M1.02	Build logical expressions using logic gates	3	Applying
M1.03	Simplify the Boolean expressions using K map and logic rules	4	Applying
Contents: Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1's complement and 2's complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De-Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K-map-advantages and disadvantages of K- Map- exercise problems			
CO2	Design the Combinational circuits such as full adder, full subtractor, Multiplexer, Gray code converter		
M2.01	Design the full adder circuit using universal gates	2	Applying
M2.02	Design the full subtractor circuit using universal gates	2	Applying
M2.03	Design the multiplexer and de multiplexer circuit using universal gates	3	Applying
M2.04	Design the BCD to decimal decoder	2	Applying

M2.05	Design the Gray to Binary code converter using universal gates	2	Applying
M2.06	Design the Binary to Gray code converter using universal gates	2	Applying
	Series Test – I	1	
Contents: Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems			
CO3	Design the Sequential logic circuits such as Asynchronous and Synchronous counter		
M3.01	Distinguish between synchronous and asynchronous sequential logic circuits	1	Understanding
M3.02	Explain SR, JK , T D, flip-flop using NAND with the help of truth table	2	Understanding
M3.03	Explain the working of master slave JK flip flop using NAND gates	2	Understanding
M3.04	Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in serial-out and serial- in parallel-out using flip flops	2	Understanding
M3.05	Explain the working of ring counter, Johnson counter using flip flops	2	Understanding
M3.06	Design asynchronous mod-N counter	1	Applying
M3.07	Design mod-8 synchronous counter	2	Applying
M3.08	Design 3-bit up down counter	2	Applying
Contents: Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8 synchronous counter, 3-bit up down counter			
CO4	Describe the working of ADC and DAC		
M4.01	Describe the working of binary weighted resistor DAC	1	Understanding
M4.02	Describe the working of R-2R ladder type DAC	1	Understanding

M4.03	Describe the working of various Analog to Digital Converter	2	Understanding
M4.04	List the specifications of DAC and ADC	1	Understanding
	Series Test – II	1	
Contents:- DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resistor method and R-2R ladder type - ADC - counter type - successive approximation type			

Text / Reference:

T/R	Book Title/Author
R1	Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, Eleventh edition – Global Edition,
R2	Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley publications, first edition.
R3	Fundamentals of Digital circuits, Anandkumar. A ,PHI
R4	Digital principles and applications. Donald P Leach, Albert Paul Malvino, Goutam Saha, McGraw Hill Publisher, 7th edition
R5	Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th edition.

Online Resources:

Sl.No	Website Link
1	https://www.digitalelectronicsdeeds.com/
2	https://www.electrical4u.com/digital-electronics/
3	https://www.tutorialspoint.com/digital_circuits/index.htm